

Experiment Nr. 31

Superconductivity

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Group Nr. M602

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1 Introduction

In 1902, Lord Kelvin raised a question that would shape low-temperature physics for the next decade: what happens to the electrical resistance of a metal as temperature approaches absolute zero? Would the reduced thermal motion of lattice ions eventually allow electrons to pass without scattering, driving resistance to zero, or would the electrons themselves freeze, sending resistance toward infinity?^[1] Neither outcome could be tested without access to temperatures far below anything achievable at the time.

This changed on 10 July 1908, when Kamerlingh Onnes successfully liquefied helium, opening up temperatures below 4 K for the first time^[1]. After early attempts with platinum were hampered by instrumentation limitations, mercury was selected as the test material due to the high purity achievable by distillation. It was later found that purity played no significant role in the onset of superconductivity, as the resistance vanished identically in pure mercury and impure amalgams alike, with lead and tin also found to be superconductors shortly after, at transition temperatures near 6.2 K and 4.2 K respectively. On 8 April 1911, cooling mercury below 4.2 K caused its resistance to drop so sharply it fell below the detection limit of the apparatus entirely^[1:2].

To quantify the effect, Onnes induced persistent currents in closed superconducting rings and monitored them over time. Any finite resistance R would cause exponential decay as

$$I(t) = I_0 e^{-\frac{R}{L}t} \tag{1}$$

No such decay was observed, bounding the resistance to at least 14 orders of magnitude below that of copper at 300 K, a difference comparable to that between a conductor and an insulator^[2]. In 1933, Meissner and Ochsenfeld further established that superconductors expel magnetic flux entirely from their interior upon cooling below T_c , identifying ideal diamagnetism as a second and independent defining property of the superconducting state^[2].

These two phenomena form the experimental basis of this report, in which the superconducting transition of $\text{YBa}_2\text{Cu}_3\text{O}_7$ sample is characterised through resistive four-point measurements and inductive AC-susceptibility measurements, alongside a demonstration of the Meissner-Ochsenfeld effect through magnetic levitation^[3].

2 Theoretical Framework

Superconductivity is characterised by the complete vanishing of electrical resistance below a material-specific critical temperature, T_c , at which point a persistent current can flow indefinitely without dissipation. Unlike an ordinary metallic conductor, whose resistance decreases only gradually as temperature is lowered, a superconductor undergoes an abrupt drop to zero resistance at T_c (see Figure 1.1). The mechanisms governing electrical transport differ fundamentally between metals, semiconductors, and superconductors, and each class exhibits a qualitatively distinct $R(T)$ behaviour. The microscopic basis for superconducting behaviour is provided by BCS theory (Bardeen, Cooper, and Schrieffer, 1957), which describes the formation of Cooper pairs through phonon-mediated electron – electron attractions, producing a dissipation-free quantum superfluid that carries current without resistance^[4].

3 Electrical Resistance and Temperature Dependence

3.1 Metals

In a metal, electrical conduction is carried by a gas of free electrons occupying states up to the Fermi energy E_F . In the Drude model, a constant electric field accelerates electrons between collisions, and the resulting steady-state current density gives the conductivity^[5]

$$\sigma = \frac{ne^2\tau}{m_e} \quad (2)$$

where n is the free electron density, e is the electron charge, m_e is the electron mass, and τ is the mean time between scattering events. The electrical resistivity follows as

$$\rho = 1/\sigma = m_e/ne^2\tau \quad (3)$$

Two scattering mechanisms contribute independently to the total resistivity^[5]. The first arises from thermally excited lattice vibrations (phonons), which become more pronounced with increasing temperature. The second comes from static crystal imperfections such as impurities and vacancies, which are temperature-independent. At high temperatures $\rho_L \propto T$, where ρ_L is the phonon contribution. So resistance increases linearly. As $T \rightarrow 0$, phonon scattering freezes out.

Figure 1: Schematic $R(T)$ dependence for a typical metal, showing approximately linear behaviour at high temperatures and saturation toward a residual resistance ρ_i at low temperatures. Adapted from Kittel^[5].

3.2 Semiconductors

In a semiconductor, the valence and conduction bands are separated by an energy gap E_g of order 1 eV^[5]. At zero temperature no free carriers exist and the material is effectively insulating. As temperature increases, electrons are thermally excited across the gap, generating mobile electrons in the conduction band and holes in the valence band, both of which contribute to conduction.

The intrinsic carrier concentration grows exponentially with temperature as

$$n_i \propto T^{3/2} \exp\left(\frac{-E_g}{2k_B T}\right) \quad (4)$$

and since this exponential factor dominates, the resistivity decreases with increasing temperature^[5]

$$\rho(T) \propto \exp\left(\frac{E_g}{2k_B T}\right) \quad (5)$$

This is the opposite behaviour to metals. A plot of $\ln \rho$ against $1/T$ yields a straight line with slope $E_g/2k_B$, providing a direct way to extract the band gap from resistance measurements^[5].

Figure 2: Schematic $R(T)$ dependence for an intrinsic semiconductor, showing exponentially decreasing resistance with increasing temperature. Adapted from Kittel^[5].

3.3 Superconductors and BCS Theory

Above T_c a superconductor behaves as a normal metal. Below T_c , however, the resistance drops abruptly to zero, which cannot be explained by reduced phonon scattering alone. The Bardeen-Cooper-Schrieffer (BCS) theory (1957) provides the microscopic explanation: a phonon-mediated attractive interaction between electrons near the Fermi surface causes them to form bound pairs known as Cooper pairs^[6]. Each Cooper pair consists of two electrons with equal and opposite momenta and opposite spins. The role of phonons in this mechanism was confirmed by the isotope effect, in which $T_c \propto M^{-1/2}$ for different isotopes of the same element^[6].

All Cooper pairs condense into the same macroscopic quantum ground state, described by a single wave function with a well-defined phase^[2]. This condensate is separated from excited states by an energy gap Δ . A thermal fluctuation cannot scatter the condensate unless it supplies enough energy to break a Cooper pair, which requires $k_B T \sim \Delta$. Below T_c this condition is not met, so scattering is suppressed entirely and resistance vanishes^[6].

The BCS theory predicts a universal relation between the zero-temperature gap and the critical temperature^[6]

$$2\Delta(0) = 3.52 k_B T_c \quad (6)$$

Experimental values across many materials typically fall in the range $3.0 k_B T_c$ to $4.5 k_B T_c$, in good agreement with the BCS prediction^[6].

Near T_c the gap closes continuously following $\Delta(T) \approx 1.74 \Delta(0)(1 - T/T_c)^{1/2}$, characteristic of a second-order phase transition^[6]. In practice, real samples such as $\text{YBa}_2\text{Cu}_3\text{O}_7$ and $\text{Bi}_2\text{Sr}_2\text{CaCu}_2\text{O}_{8+x}$ show a finite transition width ΔT_c between the onset temperature $T_{c,\text{on}}$ and the zero-resistance temperature $T_{c,0}$, attributed to inhomogeneities in oxygen concentration and crystal quality^[3].

Figure 3: Schematic $R(T)$ for a superconductor showing metallic behaviour above T_c and a sharp transition to zero resistance below it. The characteristic temperatures $T_{c,\text{on}}$, $T_{c,\text{mid}}$, and $T_{c,0}$ and the transition width ΔT_c are indicated. Adapted from^[3].

4 Magnetic Properties of Superconductors

4.1 Ideal Diamagnetism and the Meissner Effect

A superconductor cooled below its critical temperature T_c exhibits perfect diamagnetism, spontaneously expelling all internal magnetic flux in what is known as the Meissner-Ochsenfeld effect, first observed by Meissner and Ochsenfeld in 1933^[2]. The external field does not vanish abruptly at the surface but instead penetrates a thin surface layer over a characteristic length scale known as the London penetration depth, defined as

$$\lambda_L = \sqrt{\frac{m}{\mu_0 q^2 n_s}}, \quad (7)$$

where m is the carrier mass, q the charge, n_s the superfluid density, and μ_0 the permeability of free space. Beyond this surface layer the field decays exponentially as

$$B(x) = B(0) \exp\left(-\frac{x}{\lambda_L}\right), \quad (8)$$

becoming negligibly small deep within the bulk. It is worth noting that this behaviour is fundamentally different from that of a perfect classical conductor, and can only be understood within a quantum mechanical framework^[2;6].

Figure 4: Exponential decay of the magnetic field $B(x)$ within a superconductor over the London penetration depth λ_L .

4.2 Type-I vs. Type-II Superconductors

Type-I superconductors are characterised by a single critical field B_c , below which ideal diamagnetism is maintained and above which superconductivity is abruptly destroyed; elemental superconductors such as mercury and lead fall into this category^[2;6]. Type-II superconductors instead exhibit two critical fields, B_{c1} and B_{c2} , and are distinguished by a Ginzburg-Landau parameter

$$\kappa = \frac{\lambda_L}{\xi_{GL}} > \frac{1}{\sqrt{2}}, \quad (9)$$

where ξ_{GL} is the Ginzburg-Landau coherence length, which gives rise to a negative surface energy between normal and superconducting regions^[7;6]. Between B_{c1} and B_{c2} , the material enters the Shubnikov phase, in which magnetic flux partially penetrates the sample as discrete flux lines while zero electrical resistance is maintained^[2]. Notably, virtually all superconducting alloys and compounds discovered since the 1960s, including all known high-temperature superconductors where B_{c2} can exceed 100 T, belong to the Type-II class^[7;2].

4.3 Flux Quantization and the Abrikosov Vortex Lattice

In the Shubnikov phase, magnetic flux does not penetrate uniformly but is instead confined to discrete quantised tubes, each carrying exactly one magnetic flux quantum

$$\Phi_0 = \frac{h}{2e} \approx 2.07 \times 10^{-15} \text{ T m}^2, \quad (10)$$

as confirmed experimentally by Doll and Nabauer, and independently by Deaver and Fairbank, in 1961^[2]. Each vortex consists of a normal-state core of radius ξ_{GL} , which physically

Figure 5: Schematic phase diagram showing the distinction between Type-I (single critical field B_c) and Type-II superconductors (two critical fields B_{c1} and B_{c2}).

represents the spatial extent of a Cooper pair, surrounded by circulating supercurrents that decay over the length scale λ_L ^[6]. The lower and upper critical fields can be estimated as

$$B_{c1} \approx \frac{\Phi_0}{\pi\lambda_L^2}, \quad B_{c2} \approx \frac{\Phi_0}{\pi\xi_{GL}^2}, \quad (11)$$

showing that a smaller coherence length gives rise to a substantially larger upper critical field. Using Ginzburg-Landau theory, Abrikosov showed that these vortices arrange into a periodic triangular lattice that minimises the free energy of the mixed state, a prediction later confirmed through neutron diffraction and iron-colloid decoration experiments^[7;2].

Figure 6: Schematic of the Abrikosov vortex lattice in the Shubnikov phase, showing the triangular arrangement of flux tubes each carrying Φ_0 .

5 Magnetic Susceptibility and Magnetization

5.1 Diamagnetic, Paramagnetic, and Ferromagnetic Materials

5.2 Magnetization Curves of Superconductors

5.3 Critical Field, Temperature, and Current Density

6 Experimental Methods

6.1 Four-Point Probe Measurements

6.2 Inductive AC-Susceptibility Measurements

6.3 Deep-Stick Setup and Lock-In Technique

7 Conclusion

References

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